Experiment 5

Hadoop Installation, Configuration, and HDFS Operations

Subject: Data Science Lab | Environment: Single-node Hadoop | Mapped CO(s): CO2

1. Aim

To install and configure Hadoop in a single-node environment and perform basic HDFS operations using command-line tools.

2. Key Concepts

Apache Hadoop

Open-source framework for reliable, scalable storage and processing of large datasets across machines.

HDFS (Hadoop Distributed File System)

Storage layer. Files are split into blocks and stored on DataNodes. The NameNode keeps the metadata (what file lives where).

NameNode

Master daemon. Stores the file-system namespace and block locations. It does not store the actual file data.

DataNode

Worker daemon. Stores the actual data blocks and reports to the NameNode.

YARN

Yet Another Resource Negotiator. Manages cluster CPU/memory and schedules jobs.

MapReduce

Processing model: Map produces key-value pairs, Reduce aggregates them in parallel.

Replication

Each block is copied to more than one DataNode so data survives disk/node failure. Single-node labs use replication = 1.

Pseudo-distributed / single-node mode

NameNode and DataNode run on the same machine. Same commands as a real cluster, smaller setup.

3. Important Configuration Files

File	Purpose
hadoop-env.sh	JAVA_HOME and environment
core-site.xml	Default file system URI
hdfs-site.xml	Replication, NameNode/DataNode dirs
mapred-site.xml	MapReduce framework (optional here)
yarn-site.xml	YARN resource manager (optional here)

core-site.xml

fs.defaultFS = hdfs://localhost:9000

hdfs-site.xml

- dfs.replication = 1
- dfs.namenode.name.dir = local NameNode metadata folder
- dfs.datanode.data.dir = local DataNode block folder

4. Algorithm (Single-Node Setup)

- 1. Install JDK 8+ and Hadoop.
- 2. Set JAVA_HOME, HADOOP_HOME, HADOOP_CONF_DIR, and PATH.
- 3. Configure core-site.xml and hdfs-site.xml.
- 4. Format the NameNode (once): hdfs namenode -format
- 5. Start daemons: hdfs --daemon start namenode ; hdfs --daemon start datanode
- 6. Verify with jps and hdfs dfsadmin -report.
- 7. Create directories, upload, list, cat, copy, download, delete.

5. HDFS Command Cheat Sheet

Command	What it does
hadoop version	Show Hadoop version
hdfs namenode -format	Initialize NameNode metadata
hdfs --daemon start namenode	Start NameNode
hdfs --daemon start datanode	Start DataNode
jps	List Java daemons
hdfs dfs -mkdir -p /path	Create directory
hdfs dfs -put local hdfs	Upload file
hdfs dfs -ls /path	List files
hdfs dfs -cat /path/file	Print file contents
hdfs dfs -cp src dest	Copy inside HDFS
hdfs dfs -get hdfs local	Download to local disk
hdfs dfs -rm /path/file	Delete file
hdfs dfs -rm -r /path	Delete directory
hdfs dfs -df -h /	Show HDFS space
hdfs dfsadmin -report	Cluster / DataNode report

6. Exercise Answers

Exercise 1: New directory + multiple files

hdfs dfs -mkdir -p /user/lab/exercise1

hdfs dfs -put file1.txt file2.txt /user/lab/exercise1/

hdfs dfs -ls /user/lab/exercise1

Exercise 2: Copy file between HDFS directories

hdfs dfs -cp /user/lab/exercise1/file1.txt /user/lab/exercise2/file1_copied.txt

Exercise 3: Delete a directory

hdfs dfs -rm -r /user/lab/to_delete

Exercise 4: Display file contents

hdfs dfs -cat /user/lab/exercise1/file2.txt

Exercise 5: Check available space

hdfs dfs -df -h /

hdfs dfsadmin -report

7. Viva Q&A

Q: What is Hadoop?

A: Open-source framework for distributed storage (HDFS) and processing (MapReduce/YARN) of big data.

Q: What is HDFS?

A: Hadoop's distributed file system. Files are split into blocks and stored on DataNodes.

Q: NameNode vs DataNode?

A: NameNode stores metadata (namespace). DataNode stores actual blocks.

Q: What is a block in HDFS?

A: Fixed-size chunk of a file (default 128 MB in Hadoop 2/3). Last block can be smaller.

Q: Why replication?

A: Fault tolerance. If one DataNode dies, copies on other nodes keep the file available.

Q: Default replication factor?

A: 3 in a real cluster. This lab uses 1 because there is only one node.

Q: What does formatting the NameNode do?

A: Creates an empty file-system namespace. It wipes previous HDFS metadata.

Q: fs.defaultFS?

A: Property in core-site.xml that tells Hadoop where the file system lives, e.g. hdfs://localhost:9000.

Q: put vs get?

A: put uploads local -> HDFS. get downloads HDFS -> local.

Q: mkdir -p?

A: Create the directory and any missing parent directories.

Q: How do you check daemons?

A: jps should show NameNode and DataNode.

Q: HDFS vs local FS?

A: Local FS is one disk on one machine. HDFS is designed for many machines, large files, and replication.

Q: Can you edit a file in HDFS?

A: HDFS is write-once, read-many. You typically replace a file, not edit in place.

Q: What is YARN?

A: Resource manager that allocates CPU/memory and runs applications on the cluster.

Q: Single-node vs cluster?

A: Same software and commands. Cluster has many DataNodes; single-node runs both roles locally.

8. One-Line Definitions

HDFS

Distributed storage layer of Hadoop

NameNode

Master that stores file-system metadata

DataNode

Worker that stores data blocks

Block

Fixed-size piece of an HDFS file

Replication

Keeping extra copies of each block

YARN

Cluster resource manager and scheduler

MapReduce

Parallel map-then-reduce processing model

fs.defaultFS

URI of the default file system

hadoop fs / hdfs dfs

Shell for HDFS file operations

jps

Lists running Java processes / daemons